

Santhosh Kumar Ravi Kumar

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LinkedIn |Portfolio

EDUCATION

Technical University of Munich
M.Sc. in Biomedical Engineering and Medical Physics **|** June 2024 **|** Munich, Germany **|** German Grade: 2.0

National Institute of Technology, Raipur, India
B. Tech in Biomedical Engineering **|** June 2021 **|** Raipur, India **|** German Equivalent Grade: 2.1

National Institute of Technology, Warangal, India
PG Diploma in Artificial Intelligence and Machine Learning **|** March 2021 **|** Warangal, India **|** German Equivalent Grade: 1.3

Experience and Projects

(Mar 2025 – Present)

Research Scientist (AI & Software) | Ferdinand Braun Institute, Berlin:

- Developed an end-to-end AI-powered inspection platform for semiconductor laser diode facets, integrating Faster R-CNN (ResNext backbone) for multi-class defect detection and automated quality assessment.
- Engineered a PyQt5-based preprocessing application automating edge detection, diode ROI extraction, image alignment, project organization, and ML inference workflows.
- Built an interactive Flask-based inspection and QC platform using OpenSeadragon and DeepZoom for high-resolution microscopy visualization, defect review, manual QC, prediction overlays, and project-level inspection management.
- Implemented embedding-based analysis and AI-assisted annotation workflows to support intelligent sample selection, defect reclassification, and iterative model improvement, bridging ML development with scientific quality-control processes.

(Jun 2024 – Sep 2024)

Data Science Intern | **Boehringer Ingelheim:**

- Engineered and integrated advanced NLP pipelines, Python tooling, and large language Models (LLMs) to extract structured information, including tables, from clinical trial reports, automating multi-page document extraction and reducing manual effort from 6 hours to 30 minutes.
- Developed an online application for PDF uploads, result visualization, and data export in multiple formats, enhancing data extraction efficiency and accuracy, thereby optimizing research workflows.
- Implemented chat-based LLM interfaces for interactive information extraction, enabling scientists to perform faster and more effective data analysis.

(Oct 2022 – May 2024)

Software Developer | **Helmholtz Munich:**

- Designed and implemented Calimera, a visualization platform for the Computational Discovery group at the Institute of Diabetes and Obesity, enabling statistical and ML analysis of indirect calorimetry data from mouse studies.
- Built the platform using Python, Qt Designer, and Shiny for Python, allowing researchers to efficiently explore complex metabolic datasets and support faster research iteration.
- Streamlined scientific data analysis and visualization workflows, enabling researchers to explore indirect calorimetry datasets more efficiently.

PathoAICockpit | **TUM Data Innovation Lab:**
Developed a web-based platform for real-time processing and quality control of digital pathology images using CNN models in TensorFlow and supporting ML pipelines in scikit-learn. Implemented artifact detection, classification, and prediction algorithms, improving diagnostic reliability and processing efficiency in whole-slide image analysis.

Masters Thesis: Autoencoder for Disease Prediction in Mice Models (TUM & Helmholtz):
Developed an RNN/LSTM-based autoencoder to analyze temporal indirect calorimetry data from mouse models, enabling disease prediction in knockout cohorts using reconstruction loss. Applied representation learning to mitigate batch effects and improve classification performance, demonstrating the potential of deep learning for early disease detection in biomedical research.

Artificial Intelligence and Machine Learning (PG Diploma):
Gained hands-on experience in Python-based data analysis and visualization using NumPy, Pandas, and Matplotlib. Implemented supervised, unsupervised, and reinforcement learning pipelines, and developed deep learning models including CNNs, RNNs, and GANs for text classification, forecasting, and image recognition tasks.

Additional Courses

- Electrical Computer Aided Design (ECAD), MATLAB, C++, Embedded development, SDLC, Requirements Engineering (IREB), Power BI, MySQL

SKILLS

Programming & Development:	Python, C++, MATLAB, JavaScript (Angular, Node.js), Qt Designer, Shiny, Flask, Docker, Git, REST APIs, CI/CD pipelines, TensorFlow.js, High-Performance Computing (HPC), Apache Web Server (XAMPP), Embedded Systems, Agile, Scrum
Data Science & Machine Learning:	Statistical analysis, data visualization (Matplotlib, Seaborn, Power BI), Pandas, NumPy, SciPy, R, MySQL, machine learning (scikit-learn, XGBoost), deep learning (TensorFlow, PyTorch, Keras), computer vision (OpenCV, scikit-image), generative models, reinforcement learning, dimensionality reduction, supervised & unsupervised learning
Biomedical Engineering & Medical Imaging:	MRI, CT, PET, SPECT, optical and nuclear imaging, microscopy, image reconstruction algorithms, signal processing, biomedical instrumentation, biomechanics, tissue engineering (cell culture, scaffold design), device calibration and maintenance, regulatory compliance, quality assurance
Tools & DevOps:	Label Studio, OpenSeadragon, GitHub, Docker, HPC environments, testing & deployment pipelines, JIRA, Notion, Confluence
Project Management & Technical Support:	SDLC, requirements engineering (IREB), Agile & Scrum, stakeholder communication, project coordination, roadmap planning, resource allocation, multidisciplinary collaboration, technical documentation
Language Skills	English (C2), German (B1), Tamil (Native), Hindi (Fluent), Telugu (Fluent)

Positions of Responsibility

Core Member (TEDx NIT Raipur)

- Led a team of six executives to secure INR 200,000 in sponsorships over 90 days.
- Collaborated with cross-functional organizing teams to coordinate the successful execution of TEDx NIT Raipur.

Convenor (RobotiX Club, NIT Raipur)

- Spearheaded a team of 50+ coordinators and executives for the successful execution of RoboQuiz 2.0 on an online platform.
- Collaborated with campus ambassadors from various colleges, attracting 100+ participants nationwide.